

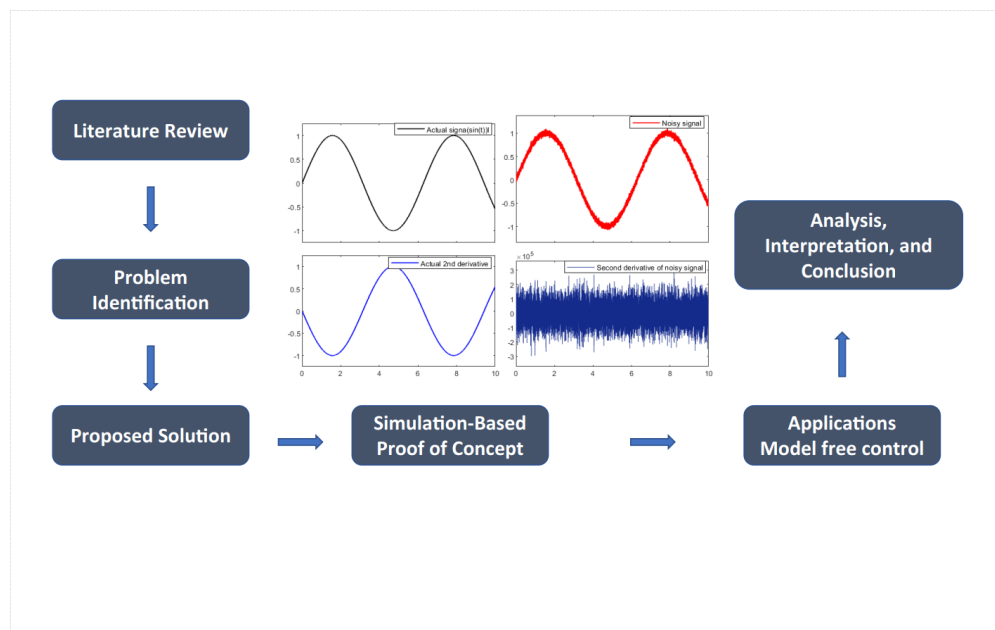
Differential methods for noisy signal differentiation: Towards a new generation of model-free controllers

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INTRODUCTION

Signal differentiation is the process of computing the derivative of a signal, which is a fundamental operation in control systems, state estimation, and signal processing. However, the estimation of derivatives from noisy signals is challenging because exact differentiation inherently amplifies high-frequency components, which include noise. This makes the derivative estimate highly unstable, leading to unreliable and erratic results, making it impractical for real-world applications. This study explores state-of-the-art differentiation methods, including high-gain observers (HGOs), high-order sliding-mode differentiators, and algebraic algorithms, to identify robust and efficient solutions for noisy environments. Two of these methods were selected for evaluation in MATLAB, where their performance was tested under different signal types and noise conditions. Key metrics such as accuracy, sensitivity to noise, computational efficiency, and stability were analyzed to determine the most effective techniques for noisy environments. Additionally, the methods were applied in a model-free control framework, yielding promising results. Finally, We proposed a modification to the High-Gain Observer (HGO) method by S. Ibrir, addressing two critical limitations of the conventional approach: the elimination of the peak phenomenon and enhanced noise robustness.

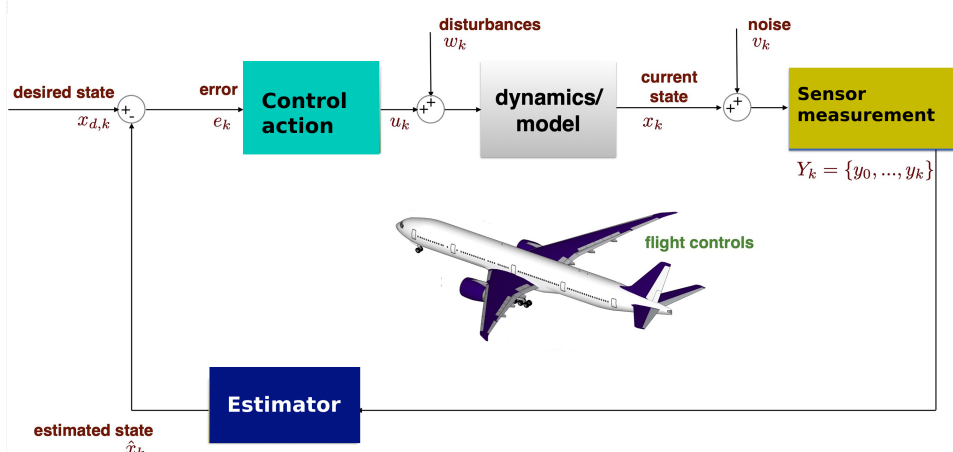
OBSERVATION/METHODOLOGY



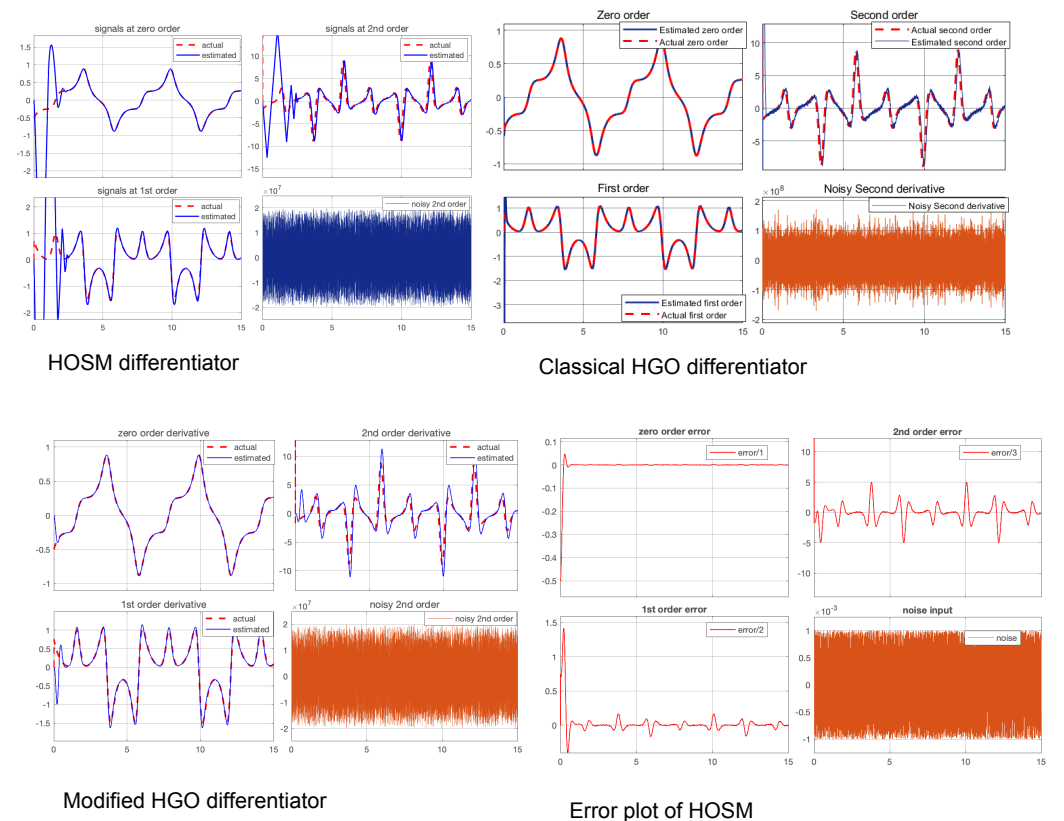
PROPOSED SOLUTIONS

- High gain observer
- High-order sliding-mode differentiators
- Numerical differentiation with annihilators
- Modified version of HGOs

Application



RESULTS



CONCLUSION & FUTURE PROSPECTS

Modified high-gain observers and sliding-mode techniques significantly improve accuracy, enhance noise suppression, and increase robustness, driving advancements in model-free control applications.

- Further research on optimizing the proposed methods
- Exploring their application in real-world scenarios, and
- Investigating their integration with advanced control techniques

REFERENCES

- [1] S. Ibrir, "Linear time-derivative trackers," Automatica, vol. 40, no. 3, pp. 397–405, 2004, ISSN: 0005-1098.
- [2] A. Levant, M. Livne, and X. Yu, "Sliding-mode-based differentiation and its application," IFAC-PapersOnLine, vol. 50, no. 1, pp. 1699–1704, 2017, 20th IFAC World Congress, ISSN: 2405-8963.